Five widely used PeMS traffic benchmarks contain unflagged imputed values: a cell-level provenance audit of PEMS-BAY, PEMS03, PEMS04, PEMS07 and PEMS08

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Abstract

The Caltrans Performance Measurement System (PeMS) records, for every detector station and 5-minute interval, how much of the reported value was measured and how much was imputed. Five widely used spatio-temporal forecasting benchmarks derived from PeMS (PEMS-BAY, PEMS03, PEMS04, PEMS07 and PEMS08) are distributed without that record. We matched every value in the released files against the PeMS station records they were drawn from. More than 99.4% of each benchmark's values are identical to PeMS (100% of those present in both sources for PEMS-BAY, PEMS04 and PEMS08; Section 4.1 gives the exceptions). **The share of benchmark values on intervals PeMS marks as fully imputed is 24.5% (PEMS-BAY), 21.6% (PEMS03), 31.8% (PEMS04), 43.5% (PEMS07) and 48.1% (PEMS08).** They are PeMS's imputed numbers, carried into the benchmarks without a flag. The four anonymized benchmarks also contain groups of nodes whose entire series are identical: 13 of 358 (PEMS03), 20 of 307 (PEMS04), 151 of 883 (PEMS07) and 25 of 170 (PEMS08) nodes. Every group traces to a PeMS station that was 100% imputed throughout the window. For the same 282 PEMS-BAY sensors still reporting today, the fully imputed share has risen from 24.5% (2017) to 54.1% (2025–26). We release the audit code so anyone with a free PeMS account can reproduce these results.

1. Background

PeMS aggregates 30-second loop-detector readings into 5-minute station records for California's state highways [1]. Missing or faulty readings are filled by an imputation procedure that uses neighboring detectors and historical patterns [2]. Each station record carries a `% observed` field: 100 means every lane in the interval was measured, 0 means the whole value was imputed, and values in between are partial.

PeMS data underlie several standard benchmarks for traffic forecasting with graph and sequence models: PEMS-BAY (speed, 325 Bay Area sensors) introduced with DCRNN [3]; PEMS04 and PEMS08 (flow, occupancy and speed) released with ASTGCN [4]; PEMS03 and PEMS07 released with STSGCN [5]; and the larger LargeST collection [6]. This report audits PEMS-BAY, PEMS03, PEMS04, PEMS07 and PEMS08. None of the five files carries per-value provenance.

2. Data

Benchmark files, taken from the authors' own distributions:

Benchmark	Source	Content
PEMS-BAY	DCRNN repository [3]; `pems-bay.h5` on the Google Drive linked from its README	speed; 325 columns = PeMS station IDs; 52,116 rows of naive local time, 2017-01-01 00:00 →

Benchmark	Source	Content
		2017-06-30 23:55 (52,128 steps minus the 12 missing at the 2017-03-12 daylight-saving change)
PEMS04	ASTGCN repository [4], `data/PEMS04/pems04.npz`	array (16992, 307, 3): flow, occupancy, speed; nodes numbered 0–306; no timestamps
PEMS08	ASTGCN repository [4], `data/PEMS08/pems08.npz`	array (17856, 170, 3); nodes numbered 0–169; no timestamps
PEMS03	ASTGNN repository [8], `data/PEMS03/PEMS03.npz` (+ `PEMS03.txt`, a list of 358 PeMS station IDs)	array (26208, 358, 1): flow; no timestamps
PEMS07	ASTGNN repository [8], `data/PEMS07/PEMS07.npz`	array (28224, 883, 1): flow; nodes numbered 0–882; no timestamps

PEMS03 and PEMS07 were introduced with STSGCN [5], whose own release is hosted on a file-sharing service we could not access. We used the copies in the repository of ASTGNN [8], by the same research group, whose PEMS04 file is byte-identical to the ASTGCN release audited here.

SHA-256 of the files audited: `pems-bay.h5`
65d69fb0a2323dba9867179eb7af47c8b814186bc459ff0a4937d21614153c8f · `pems04.npz`
95a3c9b720fffdb85f0330d09bfab41b0b3cad0ca86c0d7d5f3accacb4ac999a · `pems08.npz`
e1d03ce74e9fb79149e6e7c37c680214822c4a285baa2f42278cacabcd25c075 · `PEMS03.npz`
34859517425eded66a0014db63de54b95d0428c38890741a1710ed4d3b043de8 · `PEMS07.npz`
a07887e610aa986276209bc9384d14ed7858fdcad1ded67f53dca19eb828600b

PEMS04 and PEMS08 contain 16,992 and 17,856 five-minute steps (59 and 62 days). We used the windows most often reported for them: 2018-01-01 → 2018-02-28 (PEMS04) and 2016-07-01 → 2016-08-31 (PEMS08). Under these windows every node matches a PeMS station exactly over the full window (Section 3). We also tested 2018-07-01 → 2018-08-31 for PEMS08, which some secondary sources give: no node matches. Neither window crosses a daylight-saving change, so the absence of matches is not a time-shift artefact. For PEMS03 (91 days) and PEMS07 (98 days) we used 2018-09-01 → 2018-11-30 and 2017-05-01 → 2017-08-06; every node re-identifies under them. For PEMS03 the published station list agrees with our re-identification for 345 of 358 nodes, and the other 13 are duplicated nodes (Section 4.3) whose published and re-identified stations carry identical series.

PeMS records: Clearinghouse `station_5min` files for District 3 (2018-09-01 → 2018-11-30, 91 files), District 4 (2017-01-01 → 2017-06-30, 181 files; 2018-01-01 → 2018-02-28, 59 files), District 7 (2017-05-01 → 2017-08-06, 98 files) and District 8 (2016-07-01 → 2016-08-31, 62 files), retrieved in October 2026. Timestamps are naive Pacific local time, as in PEMS-BAY. For context we also summarize District 4 for 2025-10 → 2026-09 (363 of 365 daily files available) and District 7 (Los Angeles) for 2025-09-27 → 2026-09-25 (362 of 364).

3. Method

Matching. Each benchmark cell (timestamp, station, value) is joined to the PeMS record for the same station and timestamp. PEMS-BAY speed is compared with PeMS `avg_speed`, and PEMS04/PEMS08

flow with PeMS `total_flow`. Equality is exact; no tolerance is applied. Both sides hold the same decimal values (speed to 0.1 mph, flow as integers), so exact comparison is well defined. Each cell is classed by the PeMS record's `% observed`: measured (100), partial (1–99), imputed (0), or *no PeMS record* where PeMS has no row for that station and timestamp. Only the flow channel of PEMS04 and PEMS08 is audited (PEMS03 and PEMS07 contain flow only); occupancy and speed come from the same PeMS records.

Re-identifying nodes (PEMS03, PEMS04, PEMS07, PEMS08). For each node, candidate stations are those whose `total_flow` equals the node's flow on at least 95% of the 288 steps of any of the first three days. Each candidate is scored by the number of identical cells over the whole window, and the best score wins. Nodes with no candidate, or whose best candidate matches fewer than 99% of steps, are re-scored against every station over the whole window. Some nodes tie between several stations whose flow is identical over the entire window (Section 4.3). Our default assignment takes the lowest station ID, and we report the range of the imputed share over every possible choice among tied stations. That range has zero width for all four benchmarks to two decimal places, so the choice does not affect any reported share.

4. Results

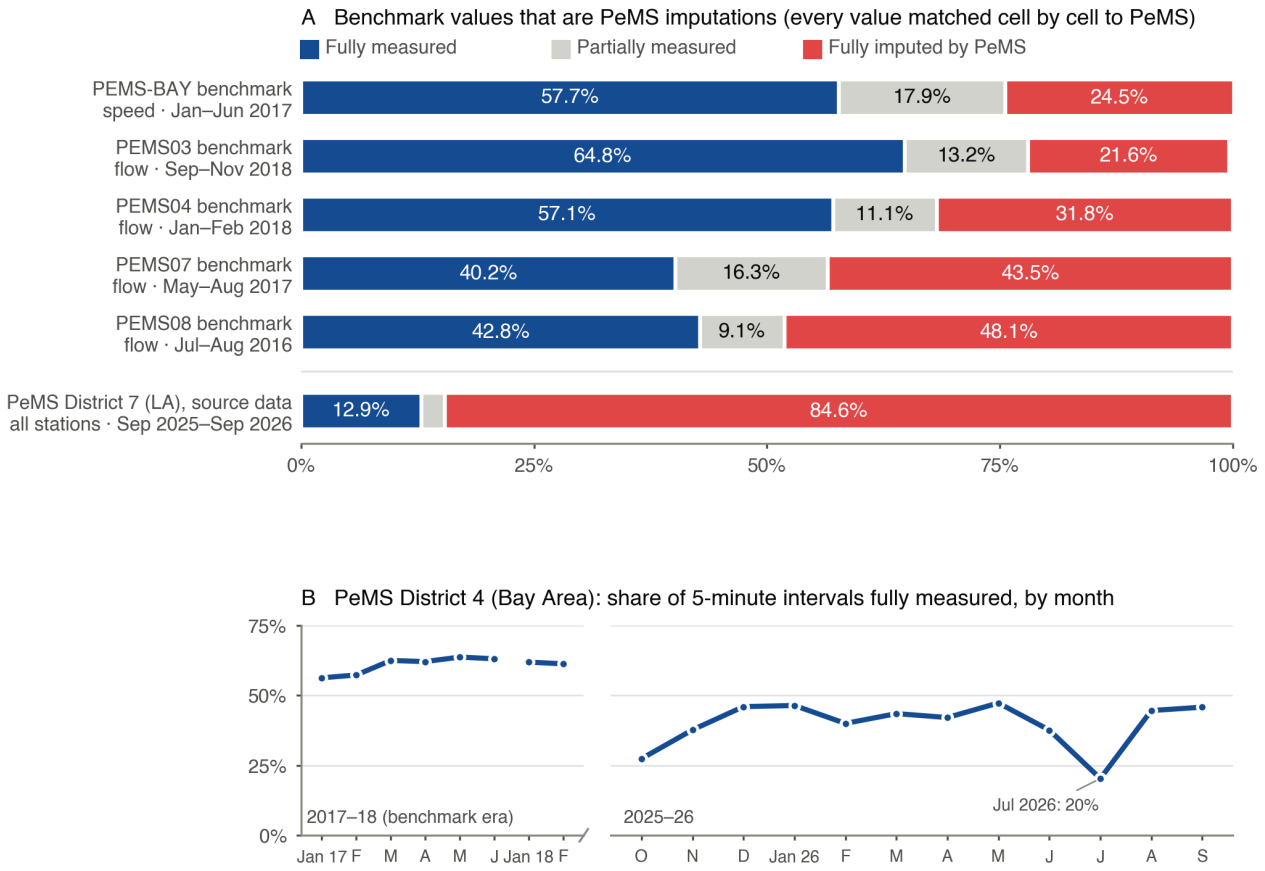


Figure 1. *(A)* Share of benchmark values on intervals PeMS marks as fully measured, partially measured or fully imputed. Benchmark values are identical to PeMS (Section 4.1), so the red segments are PeMS imputations carried into the benchmarks. The PEMS03 bar does not reach 100% because 0.4% of its cells have no PeMS record. Bottom bar: a year of current PeMS source data for Los

Angeles (District 7), for context. *(B)* Monthly share of fully measured intervals across all District 4 (Bay Area) stations, in the period PEMS-BAY and PEMS04 were drawn from and in 2025–26. The gap between periods is not drawn. Shares may not sum to exactly 100% because of rounding.

4.1 Benchmark values are PeMS values, including imputed ones

Benchmark	Cells	Fully measured	Partially measured	Fully imputed	Identical to PeMS
PEMS-BAY (speed)	16,937,700	57.7%	17.9%	**24.5%**	16,937,179 (every cell for which PeMS has a value)
PEMS04 (flow)	5,216,544	57.1%	11.1%	**31.8%**	5,216,544 (100%)
PEMS08 (flow)	3,035,520	42.8%	9.1%	**48.1%**	3,035,520 (100%)
PEMS03 (flow)	9,382,464	64.8%	13.2%	**21.6%**	9,333,445 (99.48%)
PEMS07 (flow)	24,921,792	40.2%	16.3%	**43.5%**	24,882,738 (99.84%)

PEMS03 shares are of all cells; 0.4% of its cells have no PeMS record (below).

The only PEMS-BAY cells that are not identical are 521 zeros, all where PeMS has no value. 389 fall on three 5-minute intervals for which PeMS has no record for some sensors (2017-04-20 01:40, 2017-05-05 19:55 and 2017-06-15 01:00). The other 132 are imputed intervals for which PeMS publishes no speed. The benchmark encodes these gaps as 0. DCRNN's reference implementation masks zeros (`null_val=0`) in both its loss and its evaluation metrics [3], so these cells are excluded from its scoring. The imputed values, by contrast, are non-zero and are scored.

PEMS07 behaves the same way: its 25,818 cells with no PeMS value are all zeros. Its only differing cells (13,236) all belong to a single node whose series is flat on its first days; its best station match covers 50% of the window, so that node's identification is uncertain. PEMS03 differs in one place: on 2018-09-04, 103 consecutive intervals (about 8.5 hours) are absent from the PeMS archive for every station, yet the benchmark holds values for all of them, almost all non-zero. Those 36,874 cells, and 4,120 of the 4,130 cells where both sources have a value but differ, fall on that day. The benchmark's values there cannot be traced to the current PeMS archive. They may have been filled by the authors or been present in PeMS when the data were downloaded.

4.2 Imputation is concentrated in a subset of sensors

In PEMS-BAY, 200 sensors are fully imputed on fewer than 10% of their intervals, 65 on 10–50%, 14 on 50–90%, and **46 sensors (14%) on at least 90%**. The series of those 46 sensors in the benchmark are almost entirely the output of PeMS's imputation. On 2017-01-01 and 2017-01-02 none of the 187,200 benchmark-sensor intervals is fully measured, yet both days appear complete in the released file.

4.3 The anonymized benchmarks contain duplicated, wholly imputed nodes

Some benchmark nodes have flow series that are identical to each other at every one of the window's steps:

Benchmark	Duplicated nodes	Groups (nodes each)	PeMS status of the matched station
PEMS04	20 of 307 (6.5%)	2 (15, 5)	100% imputed in every interval of the window
PEMS08	25 of 170 (14.7%)	4 (8, 8, 7, 2)	100% imputed in every interval of the window
PEMS03	13 of 358 (3.6%)	3 (8, 3, 2)	100% imputed in every interval of the window
PEMS07	151 of 883 (17.1%)	16 (31, 31, 25, 10, 9, 7, 6, 6, 6, 4, 4, 3, 3, 2, 2, 2)	100% imputed in every interval of the window

For PEMS04 and PEMS08 we checked further. These are not empty series: each has 145–326 distinct values. In PeMS the same series is carried by between 36 and 95 stations per group, so the series appears to be a common imputation profile applied to stations that measured nothing. Some duplicated nodes are directly connected in the benchmark's road graph (for example, 10 edges among the 15-node PEMS04 group), so graph models see identical neighbors.

4.4 The source has degraded since the benchmarks were built

For the same PEMS-BAY sensors, the fully measured share fell from 57.7% (Jan–Jun 2017, 325 sensors) to 35.8% (Oct 2025 – Sep 2026, the 282 sensors still reporting), and the fully imputed share rose from 24.5% to 54.1%. Across all District 4 stations, the monthly fully measured share was 56–64% in January–June 2017 and 61–62% in January–February 2018, against 20–47% between October 2025 and September 2026 (Figure 1B). In District 7 over 2025-09-27 → 2026-09-25 (4,925 stations, 509 million intervals), 12.9% of intervals were fully measured and 84.6% fully imputed. A dataset built today from these districts without provenance would carry a larger imputed share than the benchmarks above.

5. Implications

- **Evaluation targets include imputed values.** Under the standard protocols, a fifth to a half of the scored targets are PeMS imputations. Errors on those targets partly measure agreement with PeMS's imputation rather than observed traffic.
- **Inputs include imputed and duplicated values.** Models may partly learn the imputation procedure's regularities, and graph models on the anonymized benchmarks see some neighborhoods made of identical copies (in PEMS07, one node in six).
- **The effect on published results is untested.** We do not claim that any reported ranking changes. Re-scoring standard models on measured targets only is the obvious next step, and this release makes it straightforward: the class of every benchmark cell can be regenerated with the code.
- **A low-cost fix exists.** Benchmarks derived from PeMS can ship PeMS's `% observed` (or a measured/imputed mask) alongside every value, and keep station IDs so provenance stays recoverable.

6. Limitations

- PeMS may reprocess historical data. The exact equality we observe indicates that the archived values we retrieved in 2026 are the values the benchmarks contain, but we cannot rule out later changes to the `% observed` field itself.
- The PEMS03, PEMS04, PEMS07 and PEMS08 windows are inferred, not recorded in the files. The evidence for them is that every node re-identifies under them. For PEMS08 the main alternative window does not match. We did not test alternatives for the others.
- PEMS03 and PEMS07 were audited from the ASTGNN copies, not STSGCN's original release.
- One PEMS07 node (of 883) has an uncertain identification (Section 4.1). It affects that node's cells only.
- LargeST is not audited here. LargeST's published processing averages to 15-minute intervals and fills gaps with zeros [7], so PeMS-imputed values would pass through. Its audit needs different matching because the values are averages.
- The duplicated-node interpretation (a shared imputation profile) is inferred from the data; PeMS does not document it per station.

7. Reproducibility

Code: https://github.com/VLO-Labs/pems-provenance-audit (Apache-2.0). It runs on Python 3.11+ (tested on 3.12). Each audit runs in under 15 seconds on a 24-core workstation. Peak memory is about 7.5 GB, except PEMS07 (about 16 GB). The PeMS inputs total about 11 GB.

```
pip install -r requirements.txt
export PEMS_USERNAME=... PEMS_PASSWORD=...      # free PeMS account
python fetch_pems.py --district 4 --from 2017-01-01 --to 2017-06-30 --out
        pems/d04/2017
python fetch_pems.py --district 4 --from 2018-01-01 --to 2018-02-28 --out
        pems/d04/2018
python fetch_pems.py --district 8 --from 2016-07-01 --to 2016-08-31 --out
        pems/d08/2016
python fetch_pems.py --district 3 --from 2018-09-01 --to 2018-11-30 --out
        pems/d03/2018
python fetch_pems.py --district 7 --from 2017-05-01 --to 2017-08-06 --out
        pems/d07/2017
python pems_audit.py pems-bay --h5 pems-bay.h5 --pems-dir pems/d04/2017 --
        per-sensor bay_per_sensor.csv
python pems_audit.py astgcn --npz pems04.npz --start 2018-01-01 --pems-dir
        pems/d04/2018 --mapping pems04_map.csv
python pems_audit.py astgcn --npz pems08.npz --start 2016-07-01 --pems-dir
        pems/d08/2016 --mapping pems08_map.csv
python pems_audit.py astgcn --npz PEMS03.npz --start 2018-09-01 --pems-dir
        pems/d03/2018 --mapping pems03_map.csv
python pems_audit.py astgcn --npz PEMS07.npz --start 2017-05-01 --pems-dir
        pems/d07/2017 --mapping pems07_map.csv
```

We do not redistribute PeMS data or the benchmark files.

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References

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